

DWONN GOODWIN

Atlanta, GA | (404) 932-8712 | DwonnGoodwin@gmail.com | [LinkedIn](#) | dwonn.github.io

SUMMARY

Staff-level Software Engineer in Test and Automation Engineering leader with 11+ years building scalable automation frameworks, AI-native quality systems, and infrastructure-focused testing solutions for enterprise cloud applications. Known for leading cross-functional quality initiatives, mentoring engineers and SDETs, and improving release velocity by applying the test pyramid, change-aware CI/CD quality gates with affected-only test execution, and platform quality engineering practices.

EXPERIENCE

Cisco

Software Engineering Technical Leader - AI Automation & QA | August 2022 - Current

- **Architected enterprise-scale Playwright and Robot Framework** automation across six environments, supporting 1,000+ users, 850+ automated tests, 400+ API endpoints, and 40+ Lambda services, with cross-browser validation, compliance workflows, and distributed integration testing.
- **Designed CI/CD quality gates** with affected-only test execution, reducing build times by 30% while supporting bi-weekly releases across six environments through automation and defect reporting.
- **Built CI build monitoring agent** that classified failures as flakiness versus defects to stabilize pipelines, improve triage speed, and increase release confidence.
- **Built AI-driven QA orchestration workflows** that analyze PR change scope and impact, record findings in Jira for triage and routing, review QA PRs against engineering standards, generate executable E2E tests from merged PRs, and automate deployment verification.
- **Implemented adaptive merge-readiness validation** through unit/integration coverage checks, regression execution against existing E2E suites, and self-healing updates when selectors or APIs drifted.
- **Developed MCP-based integrations** across GitHub, Jenkins, AWS, Jira, and Webex to centralize multi-system automation, release validation, approval routing, and engineering workflow orchestration.
- **Delivered AI-assisted Jira tooling** that reduced manual test plan generation efforts by approximately 80% while improving workflow consistency, traceability, and structured review processes for AI-assisted engineering operations.
- **Managed AWS-based virtual test environments** using EC2, Lambda, Terraform, SQS, and synthetic test-data pipelines supporting distributed systems validation.
- **Led API validation** initiatives using pytest and Schemathesis, validating 400+ REST endpoints through automated schema checks and contract testing.
- **Built performance test and resilience** suites for concurrent load, p95/p99 latency analysis, downstream throughput validation, and database scaling.
- **Led automation strategy and quality engineering initiatives** across the team while mentoring 5 QA engineers and SDETs on automation architecture, AI-assisted workflows, and scalable testing strategies.

Amazon

Senior Software Quality Assurance Engineer | April 2021 – August 2022

- **Built cross-platform automation frameworks** for iOS, Android, and web applications using Appium and Selenium WebDriver on AWS Device Farm.
 - **Managed distributed device infrastructure**, execution scheduling, accessibility validation, and test results aggregation for large-scale cloud applications.
 - **Led QA initiatives** for a cloud music streaming platform serving millions of users, validating playback reliability, rendering behavior, latency, and platform stability across distributed environments and varying network conditions.
 - **Embedded quality earlier into the development lifecycle** through improved developer collaboration, onboarding, and automation adoption initiatives.
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Cisco

Software Test Engineer | March 2017 – April 2021

- **Designed Linux hardware-in-the-loop** and VM-based test environments for enterprise infrastructure validation, including lab configuration and multi-project system setups.
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Autotrader US

Software Test Engineer | January 2015 – March 2017

- **Developed Java/Selenium/TestNG automation** (Page Object Model, data-driven) with database and ETL validation (Oracle, SQL Server) and large-scale regression across environments.

EDUCATION

Bachelor of Business Administration in Computer Information Systems

Georgia State University

TECHNICAL SKILLS

- **Test Automation:** Playwright, Selenium WebDriver, Cypress, Python, Java, Robot Framework, Appium, pytest, TestNG, Cucumber, Schemathesis, JMeter
- **Framework Design:** Page Object Model, Hybrid, Keyword-Driven, Data-Driven; test-pyramid strategy
- **Programming:** Python, Java, JavaScript, TypeScript, Ruby, Go
- **Databases:** Oracle, SQL Server, MySQL, DynamoDB, RDS; stored procedures, schema validation
- **Cloud & DevOps:** AWS (EC2, Lambda, SQS, Device Farm, CloudWatch), Terraform, Docker, Kubernetes
- **CI/CD:** Jenkins, GitHub Actions, Git, Git Commit Hooks
- **Observability & Monitoring:** Datadog, Jira, Confluence, Tableau
- **AI Engineering & Agentic Automation:** OpenAI GPT-4 API, Model Context Protocol (MCP), NLTK, GitHub Copilot, Claude Code agent skills, Cursor
- **Collaboration:** Cross-functional Team Leadership, Agile, Sprint Planning

AWARDS AND ACKNOWLEDGEMENTS

- Recognized by Cisco leadership for leading high-impact initiatives supporting strategic engineering objectives.
- Earned an award as a top 5% performer at Amazon for showing a high level of innovation, problem-solving and initiative and for contributing as a visionary of ideas for quality.